

# Digital Signal Intelligence (DSI)

## A New Information Substrate for Insurance

### THE STRATEGIC BROKER THREAT

Brokers have always understood that their structural disadvantage is the carrier's control of risk assessment. The client relationship and data is theirs. The only thing standing between a broker and full margin ownership is the fact that risk cannot be objectively quantified without a carrier's underwriting function. Remove that dependency, and the carrier becomes a capital provider competing on appetite and price, with the broker setting the terms.

This is not a future scenario. It is the direction of sophisticated brokers already, and DSI makes it achievable. Syndicates that recognise this dynamic have two choices: own the assessment standard before brokers do or accept the margin that brokers leave them once they have.

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### TWO PROBLEMS. ONE SOLUTION

The same capability that protects syndicate margin from broker encroachment is the capability that delivers the headcount reductions already committed to investors. DSI is not an efficiency tool dressed up as strategy, it is the strategic response to an existential market shift, which also happens to solve the operational problem that currently has no viable path to resolution.

### THE TWO IMPERATIVES

#### Own the Standard First

If a syndicate embeds DSI as its assessment methodology, broker presented scores are validated against the syndicate's own standard, not substituted for it. The syndicate retains assessment authority. The broker's data enriches the process rather than replacing it. Move second, and the broker's score becomes the reference point the syndicate is asked to accept or reject on someone else's terms.

#### Defend the Margin

A broker who owns risk assessment takes no capital risk but participates in the pricing decision. That is a structurally superior margin position, risk-free influence over the premium. The syndicate's defence is not to argue against broker capability. It is to ensure that the objective standard is theirs, so that broker participation adds data to a syndicate anchored framework rather than displacing it entirely.

#### Deliver the Commitment

Syndicates have made public pledges to reduce headcount, in many cases by significant margins. The clock is running. Traditional underwriting cannot absorb those reductions without either processing fewer risks or degrading selection quality. DSI automates the assessable majority — 75–85% of Tier 1–2 risks — so the reduced headcount concentrates entirely on the risks that genuinely require judgment. It is the only credible path.

### HOW DSI DELIVERS

#### Assessment Authority

DSI scores every risk against 329 observable signals across seven categories — network authority, technical infrastructure, asset telemetry, structured data, corporate footprint, public records, and direct inquiry. The methodology is externally verifiable and grounded in the same graph-propagation principles that underpin FICO, PageRank, and modern AI. A syndicate that adopts DSI owns an assessment standard that is defensible, auditable, and impossible to dismiss.

#### Automated Majority

Tier 1–2 risks — preferred and standard — are auto-approved with full audit trail. Tier 3 auto-approves with conditions. Tier 4–5 routes to senior underwriter review. Human judgment is deployed only where it adds genuine value. The same underwriting team processes a dramatically larger book, with better average selection quality, at a fraction of current cost per bind.

## Broker Integration

Broker-submitted DSI scores are received into a syndicate framework that validates, contextualises, and may override them — not one that accepts them uncritically. The syndicate's own signal weights, tier thresholds, and conditions govern the final assessment. Broker data enriches rather than displaces. Assessment authority remains with the capital provider.

## Audit and Compliance

Every pricing decision produces a complete signal-to-tier-to-premium chain. Lloyd's Performance Management, regulatory review, and internal governance all benefit from pricing that is explainable by reference to observable inputs through a defined algorithm — not underwriter judgment that varies by individual and is difficult to interrogate retrospectively.

## THE RETROSPECTIVE PROOF

DSI's accuracy in flagging major loss events before crystallisation — SVB, FTX, MV Dali, Boeing 737 MAX — demonstrates 70–95% predictive accuracy across multiple, disparate coverage lines. In each case, the signals were observable and scoreable months to years before loss. In each case, traditional underwriting missed them. The question is not whether signal-based assessment is more accurate than self-reported documentation. It demonstrably is. The question is who owns it.

## QUESTIONS AND ANSWERS

Q: DSI sounds compelling as a methodology, but what makes it a standard and not another tool?

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The traditional underwriting model has existed for over 300 years and has never produced an accepted standard. That is a structural impossibility, not a historical accident. A model built on subjective underwriter judgment, that is neither canonical nor deterministic, cannot become a standard regardless of how long it operates or how experienced the practitioners are. The conditions for standardisation are simply absent.

DSI has the opposite characteristics. Observable signals, externally verifiable. Repeatable scoring, the same entity scored twice produces the same result. Consistent methodology, the algorithm does not vary by underwriter or syndicate. A fully auditable signal-to-tier-to-premium chain. Non-gameable inputs drawn from behaviour rather than declaration. These are the characteristics that allowed FICO to become the global credit standard, that allowed PageRank to capture 92% of search, that allow rating agency scores to function as regulatory reference points. DSI meets the same bar. The traditional model never could.

The path to full market acceptance follows a predictable sequence. A 12–24-month forward validation period — DSI-scored risks tracked against actual loss outcomes across participating syndicates — creates the independent evidence base. Once that exists, the dynamic accelerates without further persuasion. New breed capital providers, currently excluded from insurance by its opacity, enter specifically to back DSI-scored risk because the information asymmetry has been removed. Insurance becomes, for the first time, a legible asset class for yield-seeking equity. The syndicates that participate in the first validation cohort do not merely get a better underwriting tool. They become the institutions whose loss experience defines what each DSI tier means in practice — a reference position that shapes how the entire market prices risk going forward.

Q: Are brokers really moving toward owning risk assessment, or is this theoretical?

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It is directional and accelerating, not theoretical. The broker goal of reducing dependency on carrier assessment is structural and longstanding — what has changed is feasibility. Signal-based scoring methodology now exists, is validated against retrospective loss data, and is accessible via API. The brokers who move first on this will establish the standard. Syndicates that wait for the market to settle before responding will be responding on terms they did not set.

Q: We've committed to optimisations, but without a clear implementation path. How does DSI change that?

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DSI provides the only path that does not require a choice between headcount reduction and risk quality. Traditional automation reduces administrative burden but does not replace underwriting judgment, the assessment still requires a senior underwriter per risk. DSI automates the assessment itself for most risks, not just the administration around it. Tier 1–2 risks are scored, tiered, priced, and approved algorithmically. The reduced headcount handles Tier 3–5 — the risks that require experienced judgment — and handles more of them, better, than the current model allows.

Q: How does DSI interact with Lloyd's Performance Management requirements?

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DSI produces a fully auditable signal-to-premium chain that satisfies PMD's pricing transparency requirements. Every tier assignment is explainable by reference to specific observable signals through a defined algorithm. The five-tier structure maps directly to Lloyd's approved, referred, and declined workflow requirements. The audit trail — content-addressable configuration storage, full model versioning, complete signal inventory per assessment — is designed for regulatory interrogation.

Q: Does DSI support Lloyd's market structures — towers, subscriptions, co-insurance?

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Yes. DSI natively supports tower pricing via parametric ILF differentials per layer, and subscription order/line models with configurable lead loading factors. ROL validation and dual limit recommendations are built into the pricing engine for every structure. The same assessment framework that drives ground up pricing feeds tower and subscription outputs without structural change to the underwriting workflow.